American evolutionary biologist Richard Lewontin had previously suggested that animals are not passive victims of natural selection. He argued that they actively construct and modify their environment, and affect their own evolution in the process: the lynx and the hare, for example, shape each other's evolution and shared environment by striving to outrun each other. Odling-Smee similarly argued that niche construction and "ecological inheritance"—when inherited resources and conditions such as altered soil chemistry are passed on to descendants—should be seen as evolutionary processes.

"Hares do not sit around constructing lynxes! But in the most important sense, they do."

Richard Lewontin

Levels of construction

Some common examples of niche construction are obvious, while others operate at a microscopic scale. Beavers build impressive dams across rivers, creating lakes and altering river courses. This alters the composition of the water and materials carried downstream, creates new habitats for other organisms to take advantage of, and also changes the composition of the river's plant and animal communities. British biologist Kevin Laland has suggested that, while a beaver's dam is clearly of great evolutionary and ecological importance, the impact of its dung may also be significant.

Earthworms are highly effective niche constructors, constantly transforming the soil in which they live. They break down vegetable and mineral matter into particles small enough for plants to ingest. The worm casts they secrete are five times richer in usable nitrogen, have seven times the concentration of phosphates, and are about 11 times richer in potassium than the surrounding soil.